

4(a)(i)	component of momentum = $0.25 \times 3.7 \times \sin 27^\circ$	C1
	= 0.42 N s	A1
4(a)(ii)	$m_{(Z)} \times 5.5 \times \sin 44^\circ = 0.42$ or $m_{(Z)} \times 5.5 \times \sin 44^\circ = 0.25 \times 3.7 \times \sin 27^\circ$ $m_Z = 0.11 \text{ kg}$	A1
4(a)(iii)	magnitudes: equal	B1
	directions: opposite	B1
4(b)	$4 + 6 = 2 + v$ $v = 8 \text{ m s}^{-1}$	A1